

Design and Development of a Multi-Modal Deepfake Detection System Integrated into the UnMask Mobile App

Anonno Singha Ray

Department of Mechatronics Engineering
Rajshahi University of Engineering and Technology (RUET)
Rajshahi, Bangladesh
singharayanonno@gmail.com

Md. Raihanul Haque

Department of Mechatronics Engineering
Rajshahi University of Engineering and Technology (RUET)
Rajshahi, Bangladesh
raihanulhaque007@gmail.com

F. M. Rezwana Parvez

Department of Mechatronics Engineering
Rajshahi University of Engineering and Technology (RUET)
Rajshahi, Bangladesh
rezwanparvez67@gmail.com

Md. Manirul Islam

Department of Mechatronics Engineering
Rajshahi University of Engineering and Technology (RUET)
Rajshahi, Bangladesh
manirul@mte.ruet.ac.bd

Abstract—The rapid advancement of deep learning has made it possible for the generation of hyper-realistic synthetic media, commonly known as deepfakes, which pose significant threats to privacy, security, and digital trust. Most existing detection approaches available are computationally expensive and unsuitable for real-time use on mobile platforms. This paper presents UnMask, which is a lightweight and scalable multi-modal real-time deepfake detection application capable of processing image, audio, and video inputs with high accuracy. The proposed framework integrates optimized deep learning models such as DenseNet121 for image classification, a CNN-based architecture for audio analysis, and an EfficientNet-B0 with BiLSTM and attention mechanism for video detection into a FastAPI-powered backend. A corresponding lightweight mobile application, "UnMask," was developed using dart programming language and Flutter, which enables both prerecorded and live media verification using those models. Companion website was also build for the users to get to know about the app's features and download the app on mobile devices. Experimental results demonstrate strong performance, achieving accuracy scores of 99% for images and audio and 73% for video detection on benchmark datasets. These findings highlight the effectiveness of the proposed approach and its potential for real-time deployment on resource-constrained mobile devices to enhance digital media integrity.

Index Terms—Deepfake Detection, Multi-Modal System, Mobile Application, Deep Learning, CNN.

I. INTRODUCTION

The fast progress of artificial intelligence (AI) and deep learning has allowed the development of hyper-realistic synthetic media, which are also referred to as deepfakes. These fake images, produced mainly by generative adversarial networks (GANs) and analogous structures, are capable of convincingly emulating human faces, voices and even more complex gestures [1]. Although these technologies can be applied legitimately in the entertainment, education and online com-

munication sectors, they have also created significant threats to privacy, security and confidence to the masses. Deepfakes are now commonly used in disinformation campaigns, monetary fraud and identity theft, as well as cyber harassment, which threatens digital integrity and social stability [2]–[4].

These risks are increased by the increased dependency on mobile devices to consume content. Regardless of the advances in research, it is important to note that the majority of the current detection systems are still resource-intensive, cloud-based or limited to forensic and institutional applications, which makes them inappropriate to be deployed on smartphones [5], [6]. This poses a vital weakness because regular users who are the most vulnerable to deepfake content do not have readily available, real-time, verification tools. Besides, the complexity of multi-modal deepfakes, which can change audio, video and image streams at the same time, renders the old unimodal detection strategies ineffective [1], [7]. It is necessary to resolve these issues using not only the correct models of detection, but also the lightweight model, scalable, and easy to use by the non-technical population.

To resolve this challenges we have designed and developed UnMask, a lightweight and scalable multi-modal deepfake detection system for mobile deployment. The main contributions are as follows: (1) optimized deep learning models for image, audio, and video deepfake detection with low computational overhead; (2) integration of these models into a FastAPI backend for real-time inference on resource-constrained devices; (3) development of a user-friendly mobile application and companion website for broad accessibility; and (4) extensive experimental evaluation demonstrating high accuracy and generalizability across benchmark datasets.

The remainder of this paper is organized as follows. Section II reviews the related work and existing approaches in deep-

fake detection. Section III describes the methodology, including dataset acquisition, preprocessing, model development, backend integration, and application deployment. Section IV presents the experimental results and performance evaluation. Finally, Section V concludes the paper and highlights potential future directions.

II. LITERATURE REVIEW

Initial attempts at detecting a deepfake image were based on the human eye or rule-based detection, which detected artifacts, including unnatural lighting, irregular blinking, and inaccurate lip-synchronization [8]. Nonetheless, with the maturation of deepfake generation approaches, such hand methods have become more and more ineffective [9]. Deep learning has since become the paradigm. Convolutional Neural Networks (CNNs) are best at identifying visual discrepancies in images and video frames, whereas Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) models show the ability to model temporal relationships in audio and video streams [10]. Improved robustness is achieved by hybrid CNN-RNN methods which are a combination of spatial and temporal analysis [11]. More recent developments also involve attention mechanisms, which improve detection through specializing in small areas of manipulation with less computational overhead, which are also potentially useful in mobile applications [12], [13].

Adversarial training has contributed to robustness, exposing detection models to adversarially generated manipulations during training, and thus improving generalization to unseen forgeries [14], [15]. Also, multi-modal detection systems that share the analysis of visual, auditory, and temporal information have been effective in detecting cross-modal anomalies like discrepant lip movements and speech timing [14], [16].

Lightweight optimization methods, including model pruning, quantization, and knowledge distillation have become essential enablers of real-time deployment because of the constraints of mobile platforms [10], [17]. A range of optimized architectures, such as InceptionResNetV2, ResNet-50 and the hybrid CNN-LSTM models, have demonstrated detection rates of over 95% on test datasets, including FaceForensics++, Celeb-DF, and the Deepfake Detection Challenge (DFDC) [18]–[20]. There are additional new techniques, including ensemble learning based on biological signals [21], detection of lighting inconsistencies [18], and watermarking schemes [22], which enlarge the detection space. Despite these advancements, challenges remain in dataset diversity, cross-dataset generalization, and deployment on resource-constrained mobile devices. Most existing methods are either modality-specific or computationally heavy, leaving a gap in lightweight, real-time, multi-modal detection systems optimized for mobile platforms. This research seeks to bridge that gap by developing Unmask, a scalable, mobile-based solution that integrates deep learning models with a user-centric design to strengthen digital media integrity in the face of synthetic manipulation.

III. METHODOLOGY

This section outlines the datasets, preprocessing techniques, model architectures, backend integration, mobile app implementation, and website development, detailing the complete pipeline of the proposed multi-modal deepfake detection system. Fig. 1 shows the workflow diagram of our Unmask mobile application.

A. Dataset acquisition and preprocessing

For image deepfake detection, we used a dataset of 140k real and fake faces. This dataset consists of 70,000 fake faces selected from 1 million synthetic faces produced by StyleGAN which was supplied by Bojan, and also 70,000 real faces taken from the Flickr-Faces-HQ (FFHQ) dataset that was gathered by Nvidia. Training images were resized to 256×256, randomly flipped for augmentation, converted to tensors, and normalized using ImageNet statistics. Validation and test sets followed the same pipeline without flipping. Data were organized with ImageFolder and loaded via PyTorch DataLoader with shuffling (training only) and a batch size of 64 for efficient processing.

We used an open-source dataset “In the wild” for training our audio deepfake detection model. The dataset basically consist of audio deepfakes for a set of politicians and other public figures, collected from publicly available sources such as social networks and video streaming platforms. For the number of 58 celebrities and politicians, the dataset has a collection of 20.8 hours of real and 17.2 hours of fake audio. It consists of 19963 real audio samples and 11816 fake audio samples of short durations. All input audio samples were standardized to a sampling rate of 22.05 kHz with a fixed duration of 2 seconds. To ensure uniformity, shorter clips were zero-padded while longer clips were trimmed. Each waveform $x(t)$ was transformed into a Mel-spectrogram representation using the Short-Time Fourier Transform (STFT), defined as:

$$S(m, k) = \sum_{n=0}^{N-1} x[n] \cdot w[n - m] \cdot e^{-j2\pi kn/N}, \quad (1)$$

where $w[n]$ is the windowing function, m is the frame index, and k denotes the frequency bin. A Mel filter bank and logarithmic scaling were applied to enhance perceptual relevance. Fig.2 shows the Mel spectrogram of real and fake images, respectively.

Video data were collected from the FaceForensics++ dataset, excluding the official test split to prevent leakage. An 80/20 stratified division maintained class balance. From each clip, 20 evenly spaced frames were extracted using OpenCV, resized to 224×224, and padded with repeated or zero frames if necessary to ensure uniform length. For training, frames underwent random horizontal flipping and color jittering, while validation and test sets used only resizing and normalization (ImageNet mean and standard deviation). Each video was represented as a tensor of shape (20, 3, 224, 224). Data loading was handled via a custom PyTorch Dataset with a weighted sampler to address class imbalance.

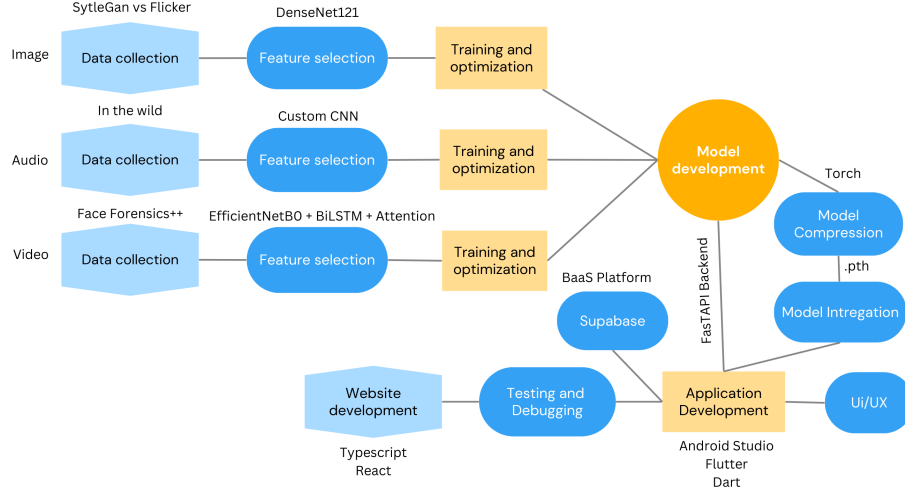


Fig. 1. Workflow diagram of Unmask

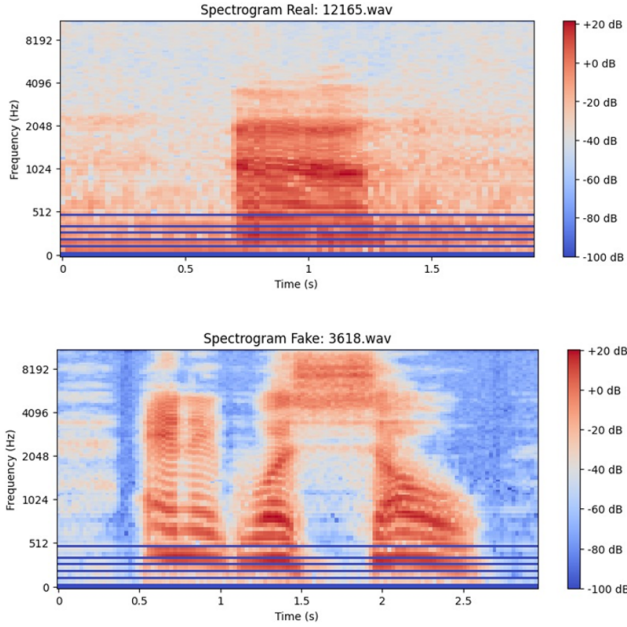


Fig. 2. Real and Fake Mel spectrogram (MFCC)

B. Model Development

1) *Deepfake Image Detection*: For image-based deepfake detection, we employed a transfer learning approach using DenseNet121 as the backbone. The final classification layer of DenseNet121 was replaced with a custom fully connected head consisting of a linear layer (1024 $\rightarrow$ 512), ReLU activation, batch normalization, dropout, and a final sigmoid output neuron. This allowed the model to map high-dimensional feature embeddings into a binary output space, distinguishing between real and fake faces.

Given an input image I , DenseNet121 extracted a feature vector x as:

$$x = \text{DenseNet121}(I) \quad (2)$$

The classification output was obtained as:

$$\hat{y} = \sigma(Wx + b) \quad (3)$$

The model was optimized using Binary Cross-Entropy (BCE) loss and the Adam optimizer with a learning rate of 0.001 was employed along with a learning rate scheduler to dynamically adjust learning rates when validation loss plateaued. Dropout with a rate of 0.3 was used for regularization.

2) *Deepfake Audio detection*: The proposed classifier is a Convolutional Neural Network (CNN) designed to capture both spectral and temporal cues from the Mel-spectrograms. The architecture consists of three convolutional blocks and two fully connected layers. The convolutional operation is defined as:

$$y_{i,j}^{(l)} = \sigma \left(\sum_{c=1}^C (x_c^{(l-1)} * w_c^{(l)})_{i,j} + b^{(l)} \right), \quad (4)$$

where $x^{(l-1)}$ is the input feature map, $w_c^{(l)}$ is the convolutional kernel, $b^{(l)}$ is the bias term, and $\sigma(\cdot)$ denotes the ReLU activation.

The network was trained using the binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (5)$$

where $y_i \in \{0, 1\}$ denotes the ground-truth label ($fake = 0$, $real = 1$), and $\hat{y}_i$ is the predicted probability. Optimization was performed using the Adam optimizer with a learning rate of 1×10^{-3} and weight decay regularization. Dropout was applied before the final classification layer to mitigate overfitting.

3) *Deepfake Video Detection*: Each video was represented by 20 uniformly sampled frames, resized to 224×224 and normalized. EfficientNet-B0 extracted spatial embeddings from each frame:

$$x_t = \text{EfficientNetB0}(F_t) \quad (6)$$

The sequence $\{x_1, x_2, \dots, x_T\}$ was passed to a BiLSTM to capture temporal dependencies. An attention mechanism assigned weights to highlight forgery-relevant frames:

$$H = \sum_{t=1}^T \alpha_t h_t, \quad \alpha_t = \frac{\exp(w^\top h_t)}{\sum_{k=1}^T \exp(w^\top h_k)} \quad (7)$$

where h_t is the BiLSTM hidden state at time t , w is a learned attention vector used to score each h_t , α_t is the normalized attention weight for time t defined by the softmax, and T is the total number of frames.

The aggregated representation H was classified using a sigmoid layer:

$$\hat{y} = \sigma(W_{fc}H + b_{fc}) \quad (8)$$

where W_{fc} and b_{fc} are the final classification weight matrix and bias. Focal loss was employed during training to address class imbalance and emphasize harder samples. This combination of CNN-based spatial features, BiLSTM temporal modeling, and attention weighting enabled accurate detection of manipulated videos.

C. FastAPI-Based Multi-Modal Backend

The UnMask system employs a FastAPI-based backend to support multi-modal deepfake detection on images, audio, and video. The service handles API requests, preprocesses inputs, forwards them to pretrained models, and returns predictions in real time. Hardware optimization enables automatic device selection (CUDA, MPS, or CPU), ensuring efficiency across platforms. The backend is started with the Uvicorn ASGI server to allow access to the device's frontend. It is basically configured to bind to all interfaces (0.0.0.0) on port 8082. It helps mobile and other devices out on the same local network to access the backend services by reaching the IP address of the host machine. Fig. 3 shows the API endpoints for client uploads (image, audio, video).

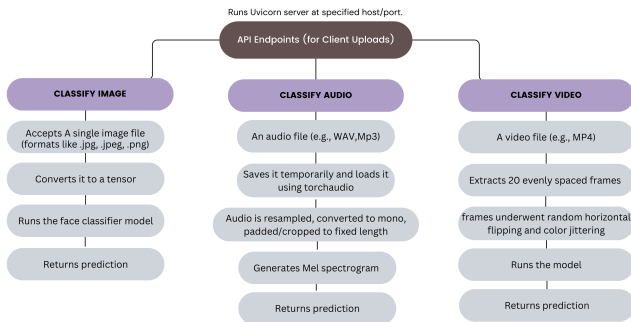


Fig. 3. API endpoints

D. Mobile App Development

The 'Unmask' app aims to provide a user-friendly and powerful tool for detecting deepfakes, which enables users to conduct media analysis on their mobile phones. We used

dart programming language and flutter-an open-source framework made by Google, for building the app. Simulation and coding was done on Android Studio. We used Supabase as the backend-as-a-service (BaaS) platform that handles user authentication, session management, and secure data storage on the backend of our app.

After uploading an image, audio, or video, the app performs rapid analysis and provides an authenticity prediction with a confidence score. This real-time feedback is crucial for users requiring immediate results. Beyond pre-recorded media, the app also supports live capture of photos, videos, and audio recordings, enabling on-the-spot verification. Fig. 4 shows the user interface of the UnMask application.

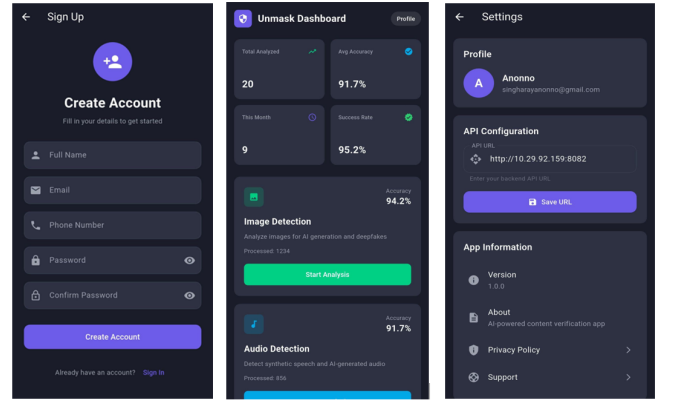


Fig. 4. User interface of the UnMask app.

E. Website Development

The main goal of our website is to create a simple yet engaging platform where users can easily learn more about our deepfake detection app, get to know about the app's features, and download the app on mobile. Fig. 5 shows the interface of the website.

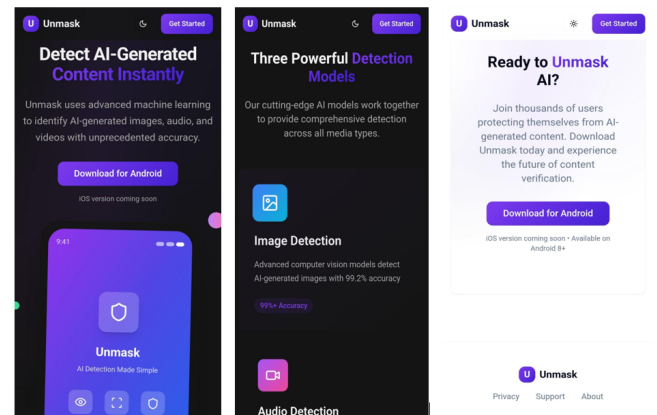


Fig. 5. Website of our system

The website has been developed using TypeScript and React, which comes with strong development tools and type safety. CSS Grid and Flexbox has been used for make content

fluid on different screen sizes. We also used Tailwind CSS for fast, consistent styling and shadcn/ui for ready-made accessible components. The website also uses day and night mode for enhanced user accessibility and comfort. It's hosted on lovable.

IV. RESULTS

The deepfake image classification model achieved strong performance, with a test accuracy of 99%, precision of 99.78%, recall of 96.42%, and an F1-score of 98.07%. Training and validation accuracy curves in Fig. 6 indicate stable learning with minimal overfitting. These results confirm the model's high reliability in distinguishing real and fake images across diverse test samples.

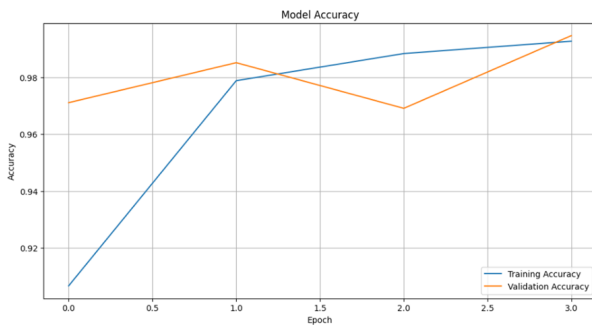


Fig. 6. Training vs validation accuracy of image deepfake detection model

Grad-CAM generates heatmaps by computing gradients of class scores with respect to convolutional feature maps, highlighting critical image regions influencing predictions, where deeper layers capture progressively important discriminative features. In Fig. 7, one of the three different convolutional layers is compared to show how deeper layers find important features. The lowest value in the "jet" colormap is deep blue, and the highest value is deep red.

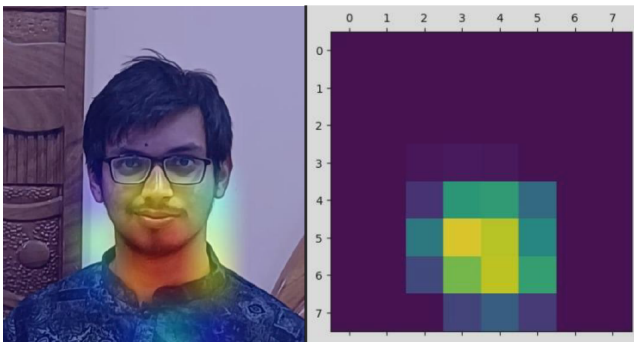


Fig. 7. GradCam Visualization

The deepfake audio classification model demonstrated excellent performance, achieving an accuracy of 99.67%, precision of 99.48%, recall of 100%, and an F1-score of 99.74%. Fig.8 shows training and validation curves that show rapid convergence with minimal loss and near-perfect accuracy,

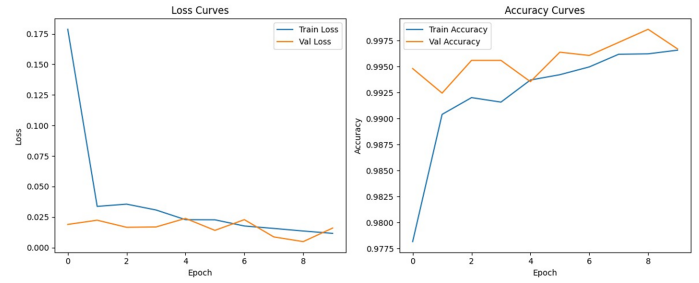


Fig. 8. Training vs validation loss and accuracy of audio deepfake detection model

TABLE I
PERFORMANCE METRICS OF DEEPAKE DETECTION MODELS

Model	Acc.	Prec.	Rec.	F1
Image	98.10	99.78	96.42	98.07
Audio	99.67	99.48	100.00	99.74
Video	73.75	82.76	60.00	69.70

confirming the model's robustness and its ability to accurately differentiate between real and fake audio samples.

The proposed EfficientNet-B0 + BiLSTM + attention model showed strong performance in deepfake video detection. The model achieved a training loss of 0.0403 with 80.0% training accuracy, while the validation loss was 0.1155 with a validation accuracy of 73.75%. The precision reached 82.76%, indicating strong reliability in correctly identifying deepfakes, although the recall was comparatively lower at 60.0%, reflecting missed detections. The area under the ROC curve (AUC) of 0.7794 further suggests a fairly robust discriminative ability, though improvements are needed for better generalization.

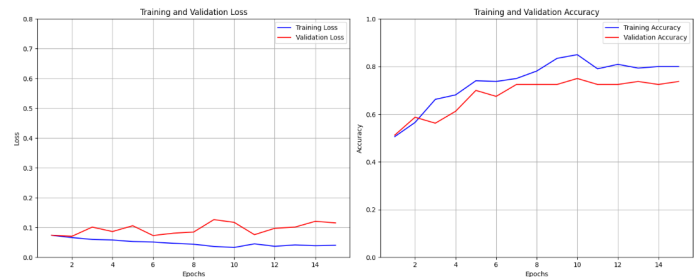


Fig. 9. Training vs validation loss and accuracy of video deepfake detection model

The confusion matrices in Fig. 10 highlight the performance of the proposed deepfake detection models across image, audio, and video modalities. The image model achieved near-perfect classification with minimal misclassifications, while the audio model demonstrated high reliability with almost no false positives. In contrast, video detection showed relatively lower accuracy due to temporal and compression artifacts. These results emphasize that while image and audio detection perform robustly, video deepfake detection requires further optimization.

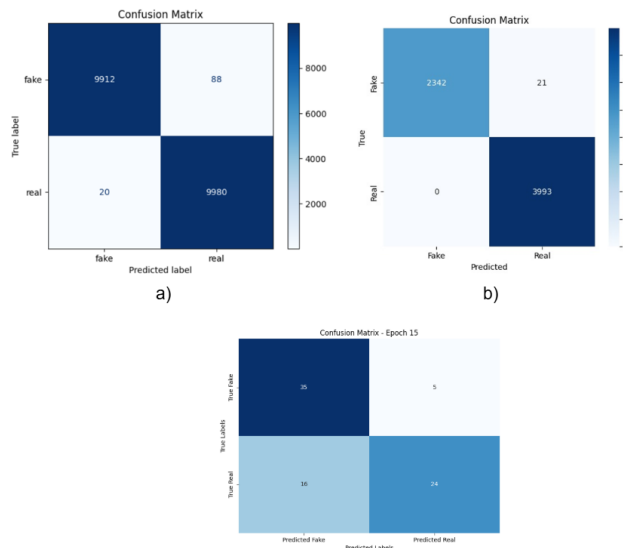


Fig. 10. Confusion matrices a) image deepfake detection b) audio deepfake detection c) video deepfake detection

V. CONCLUSION

This work introduced UnMask, a practical and scalable multi-modal deepfake detection system tailored for real-time use on mobile platforms. By combining lightweight deep learning models for image, audio, and video analysis with an efficient FastAPI backend and user-friendly interfaces, the system bridges the gap between high-performance detection and accessibility for everyday users. Experimental results revealed near-perfect accuracy for image and audio detection and competitive performance for video classification, highlighting both the strengths and areas for refinement. Looking ahead, we aim to further enhance the robustness of video detection by addressing temporal inconsistencies and improving cross-dataset generalization. Additionally, future developments will focus on expanding support for newly emerging deepfake modalities while keeping the system lightweight and user-centric. We believe that UnMask provides a strong step toward safeguarding digital trust and empowering users with accessible tools to counteract synthetic media threats.

REFERENCES

- [1] N. Chakravarty and M. Dua, "A lightweight feature extraction technique for deepfake audio detection," *Multimedia Tools and Applications*, vol. 83, no. 26, pp. 67443–67467, Jan. 2024, doi: <https://doi.org/10.1007/s11042-024-18217-9>.
- [2] F. Zafar, T. A. Khan, S. Akbar, M. T. Ubaid, S. Javaid, and K. A. Kadir, "A Hybrid Deep Learning Framework for Deepfake Detection Using Temporal and Spatial Features," *IEEE Access*, vol. 13, pp. 79560–79570, Jan. 2025, doi: <https://doi.org/10.1109/access.2025.3566008>.
- [3] B. Zhang, H. Cui, V. Nguyen, and M. Whitty, "Audio Deepfake Detection: What Has Been Achieved and What Lies Ahead," *Sensors*, vol. 25, no. 7, p. 1989, Mar. 2025, doi: <https://doi.org/10.3390/s25071989>.
- [4] H. H. Nguyen, J. Yamagishi, and I. Echizen, "Capsule-Forensics Networks for Deepfake Detection," *Advances in Computer Vision and Pattern Recognition*, pp. 275–301, Jan. 2022, doi: https://doi.org/10.1007/978-3-030-87664-7_13.
- [5] A. Malik, M. Kuribayashi, S. M. Abdullahi, and A. N. Khan, "DeepFake Detection for Human Face Images and Videos: A Survey," *IEEE Access*, vol. 10, pp. 18757–18775, Jan. 2022, doi: <https://doi.org/10.1109/access.2022.3151186>.
- [6] S. Agarwal, H. Farid, Y. Gu, M. He, and K. Nagano, "Detecting deepfake videos from phoneme-viseme mismatches," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 6596–6605.
- [7] I. Masi, Y. Wu, T. Hassner, and P. Natarajan, "Two-branch recurrent network for isolating deepfakes in videos," in *Proc. Eur. Conf. Computer Vision (ECCV)*, 2020, pp. 667–684.
- [8] A. Malik, M. Kuribayashi, S. M. Abdullahi, and A. N. Khan, "DeepFake Detection for Human Face Images and Videos: A Survey," *IEEE Access*, vol. 10, pp. 18757–18775, Jan. 2022, doi: <https://doi.org/10.1109/access.2022.3151186>.
- [9] D. Sabareshwar, S. Raghul, S. Ravi, M. Varalakshmi, and Mohamed, "A Lightweight CNN for Efficient Deepfake Detection of Low-resolution Images in Frequency Domain," pp. 1–6, Feb. 2024, doi: <https://doi.org/10.1109/ic-etite58242.2024.10493406>.
- [10] A. Khormali and J.-S. Yuan, "ADD: Attention-Based DeepFake Detection Approach," *Big Data and Cognitive Computing*, vol. 5, no. 4, p. 49, Sep. 2021, doi: <https://doi.org/10.3390/bdcc5040049>.
- [11] M. P. Kumar, V. V. Kumari, and L. Durga, "Deepfake Video Detection Using Inception Resnet V2: A Convolutional Neural Network Approach," pp. 595–601, Dec. 2024, doi: <https://doi.org/10.1109/cicn63059.2024.10847574>.
- [12] G. Chen, C. Du, Y. Yu, H. Hu, H. Duan, and H. Zhu, "A Deepfake Image Detection Method Based on a Multi-Graph Attention Network," *Electronics*, vol. 14, no. 3, p. 482, Jan. 2025, doi: <https://doi.org/10.3390/electronics14030482>.
- [13] C. Lu, B. Liu, W. Zhou, Q. Chu, and N. Yu, "Deepfake Video Detection Using 3D-Attentional Inception Convolutional Neural Network," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, Sep. 2021, pp. 3572–3576, doi: <https://doi.org/10.1109/icip42928.2021.9506381>.
- [14] S. Lei, J. Song, F. Feng, Z. Yan, and A. Wang, "Deepfake Face Detection and Adversarial Attack Defense Method Based on Multi-Feature Decision Fusion," *Applied Sciences*, vol. 15, no. 12, p. 6588, Jun. 2025, doi: <https://doi.org/10.3390/app15126588>.
- [15] S. Khan, J.-C. Chen, W.-H. Liao, and C.-S. Chen, "Adversarially Robust Deepfake Detection via Adversarial Feature Similarity Learning," *Lecture Notes in Computer Science*, pp. 503–516, 2024, doi: https://doi.org/10.1007/978-3-031-53311-2_37.
- [16] S. Waseem, A. Rahman, Z. Omar, B. A. Ahmed, S. Baloch, and A. Hafeezallah, "Multi-attention-based approach for deepfake face and expression swap detection and localization," *EURASIP J. Image and Video Processing*, vol. 2023, no. 1, Aug. 2023, doi: <https://doi.org/10.1186/s13640-023-00614-z>.
- [17] J. K. and A. M., "Safeguarding media integrity: A hybrid optimized deep feature fusion based deepfake detection in videos," *Computers Security*, vol. 142, p. 103860, Jul. 2024, doi: <https://doi.org/10.1016/j.cose.2024.103860>.
- [18] A. Qadir, R. Mahum, M. A. El-Meligy, A. E. Ragab, A. AlSalman, and H. Hassan, "An efficient deepfake video detection using robust deep learning," *Heliyon*, vol. 10, no. 5, pp. e25757, Feb. 2024, doi: <https://doi.org/10.1016/j.heliyon.2024.e25757>.
- [19] A. V. Nadimpalli and A. Rattani, "ProActive DeepFake Detection using GAN-based VisibleWatermarking," *ACM Trans. Multimedia Computing, Communications, and Applications*, vol. 20, no. 11, pp. 1–27, Sep. 2024, doi: <https://doi.org/10.1145/3625547>.
- [20] O. P. Singh, K. N. Singh, A. K. Singh, and A. K. Agrawal, "Deep learning-based image encryption techniques: Fundamentals, current trends, challenges and future directions," *Neurocomputing*, vol. 612, p. 128714, Jan. 2025, doi: <https://doi.org/10.1016/j.neucom.2024.128714>.
- [21] R. Sunil, P. Mer, A. Diwan, R. Mahadeva, and A. Sharma, "Exploring Autonomous Methods for Deepfake Detection: A Detailed Survey on Techniques and Evaluation," *Heliyon*, p. e42273, Jan. 2025, doi: <https://doi.org/10.1016/j.heliyon.2025.e42273>.
- [22] B. Guo, H. Tai, G. Luo, and Y. Zhu, "AVSecure: An Audio-Visual Watermarking Framework for Proactive Deepfake Detection," pp. 1–4, May 2024, doi: <https://doi.org/10.1109/icciec61773.2024.10561738>.